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SC2[illegible]5 Discrete Mathematics
Winter 2024-25

In-Sem 1 Examination
Duration: 90 mins

February 8, 2025
Marks 25

1. Consider a generic relation defined over the set of positive integers, which says that $(x, y) \in R$ if and only if,

$$(((x < y) \land ((y - x) \leq 200)) \lor ((x - y) \geq 150))$$

(30,60) (60,61) 60,30 (160,10)

Determine whether this relation is: [6]

(a) reflexive/irreflexive or neither

(b) symmetric/anti-symmetric or neither

(c) transitive

2. Suppose the size of a finite set $|S|$ divides the size of its power set $\mathcal{P}(S)$. What are the possible values of $|S|$? [4]

3. Answer the following questions with valid arguments. [3+2+3]

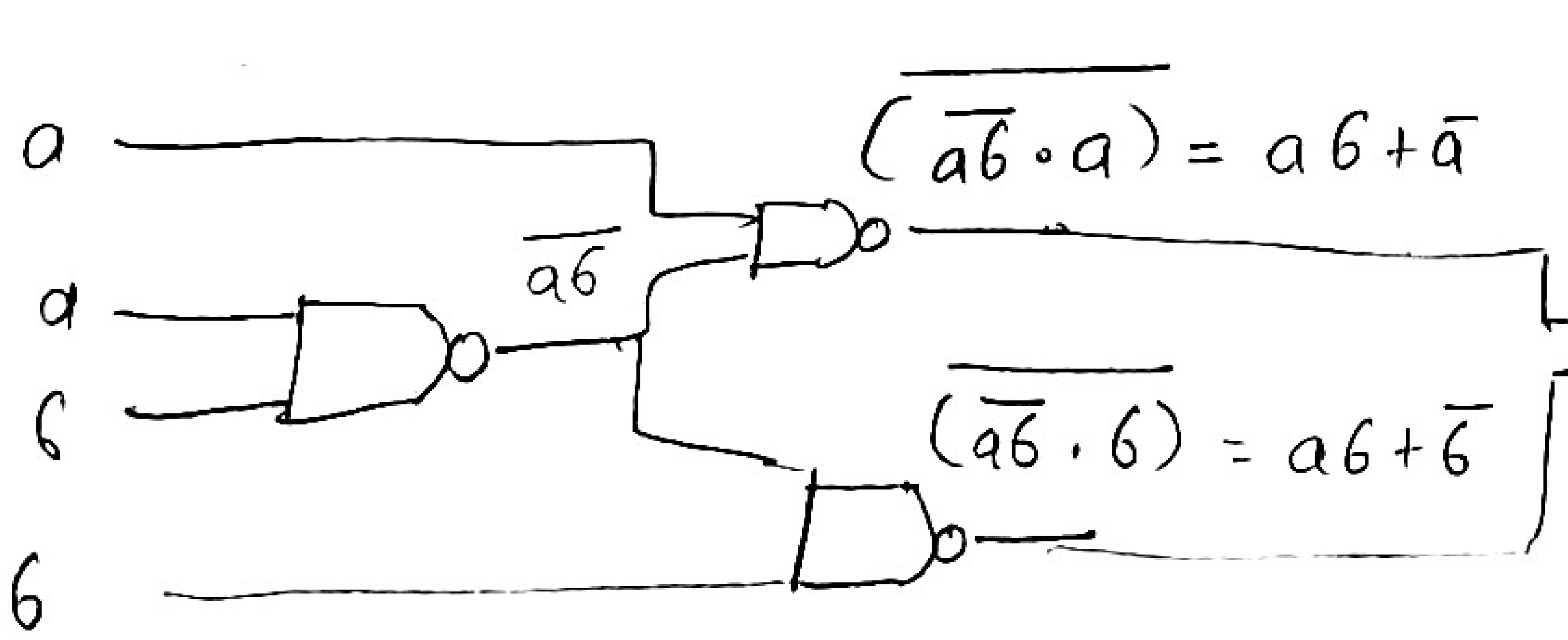
(a) Let $f : A \to B$ and $g : B \to A$ be two functions. Prove/disprove: If $g \circ f : A \to A$ is invertible, then at least one of f, g must be invertible.

(b) Let $f : A \to A$ be a function. For any positive integer $n > 1$, define the function $f^n : A \to A$ as $f^n := \underbrace{f \circ f \circ \ldots \circ f}_{n-1 \text{ compositions}} : A \to A$. Prove/disprove: If for some positive integer $n > 1$, $f^n = id$, the identity function on set A, then the function f must be bijective.

2,1 1,2 (6,6) (9,9)

(c) Let $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$ be defined as $f(x, y) = (x + 3y + 2, 2x + 5y - 3)$. Find whether f is invertible or not. Note that $\mathbb{Z}$ denotes the set of all integers.

4. Can you express the propositional logic formula $p_1 \land p_2$, using only the connectives $\oplus$ and $\neg$. If so, give the expression and show how you derive it. If not, explain why it is not possible. [7]



$\overline{(ab+\bar a)\cdot(ab+\bar b)}$

$(\bar a b+\overline{ab})$

$= (\bar a+\bar b)\cdot\overline{(\bar a\bar b)}$

$= (\bar a+\bar b)\cdot(a+b)$

$= \bar a b + a\bar b$

$= a \oplus b$

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SC205 Discrete Mathematics
Winter 2024-25

In-Sem 2 Examination — March 25, 2025
Duration: 90 mins — Marks 30

1. For the following sets and relations defined on them, find whether the relation is reflexive, anti-symmetric and transitive (and hence, a poset) or not. If it is a poset, find if it is also a lattice. **[10]**

   (a) Consider the relation S over the set $\mathbb{Z}^+$ of positive integers, where $(a,b) \in S$ if and only if $\lfloor \frac{a}{11} \rfloor \geq \lfloor \frac{b}{11} \rfloor$ and $a\%11 \leq b\%11$. reflexive, anti-symmetric,

   11 34 23 22, 23 23, 22

   (b) Consider the set A of all non-empty finite length open intervals from $\mathbb{R}$, i.e., $A = \{(a,b) \subset \mathbb{R} \mid b > a,\ a,b \in \mathbb{R}\}$. A relation S is defined on A as: For any $I_1 = (a_1,b_1) \in A, I_2 = (a_2,b_2) \in A$, $(I_1,I_2) \in S$ if and only if $|b_1 - a_1| \leq |b_2 - a_2|$. reflexive & anti-symmetric transitive

   (c) A point electric charge with charge Q C is placed at the origin of $\mathbb{R}^2$. Now, consider the set $A = \mathbb{R}^2 \setminus \{(0,0)\}$, on which the relation S is defined as: For all $(x_1,y_1) \in A, (x_2,y_2) \in A, ((x_1,y_1),(x_2,y_2)) \in S$ if the electric potential V at these points due to the charge placed at the origin satisfies $V(x_1,y_1) \leq V(x_2,y_2)$.

2. There are seven fundamental *swaras* or notes in Indian classical music: $\{$*sa, re, ga, ma, pa, dha, ni*$\}$, listed in an ascending order. **[4]**

   (a) How many distinct sequences of length 10 can you form from these *swaras*, given that repetition is allowed?

   (b) How many distinct *non-decreasing* sequences of length 10 can you form from these *swaras*, given that repetition is allowed?

3. How many ways are there to fill the entries of a 4×4 grid with the digits 1, 2, 3, 4 in such a way that no row has more than one occurrence of the same number and no column has more than one occurrence of the same number? **[8]**

4. Consider the list $1,2,3,4,5,6,7,8$. How many arrangements of this list are there, such that no element is more than two positions away from its initial position. For example the element at position 1 can end up at either position 1, 2 or 3. The element at position 8 can end up at either position 6, 7 or 8. The element at position 4 can end up at position 2, 3, 4, 5 or 6. The element at position 7 can end up at position 5, 6, 7 or 8. **[8]**

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SC205 Discrete Mathematics
Winter 2024-25

End Semester Examination
Duration: 3 hrs.

April 30, 2025
50 Marks

Note: Direct answers without appropriate justification will not be awarded any marks.

1. Consider the formulae:
   $\phi_1 = p \vee q \vee r$
   $\phi_2 = p \wedge q \wedge r$
   $\phi_3 = p \oplus q \oplus r$
   $\phi_4 = (p \Rightarrow q) \Rightarrow r$
   $\phi_5 = p \Rightarrow (q \Rightarrow r)$

   (a) Arrange the above formulae in increasing order of number of satisfying assignments (assignments that make the formula true). [3]

   (b) List the ordered pairs of a relation R over this set of propositional logic formulae, where $(a, b) \in R$ if and only if $a \Rightarrow b$. State if the relation is reflexive/irreflexive/neither; symmetric/antisymmetric/neither; transitive. [5]

2. Consider a building that has three lifts A, B, C. Lift A can take you between any pair among the floors $G, 2, 4, 6, 8, 10$; Lift B can take you between any pair among the floors $2, 3, 5, 7, 10$ and lift C can take you between any pair among the floors $1, 7, 9$. You need to deliver items gathered at ground the floor (all items are picked up at once), in increasing order of floor numbers, to floors 1 to 10. You can move between two lifts on any floor that they both service.

   (a) What is the total number of destination floor buttons you need to press (inside the lifts), to complete all the ten deliveries? [2]

   (b) Disable any one floor for any one lift to increase the number of destination button presses as much as possible, but still allowing all deliveries to take place. What is the new number of button presses needed? [3]

   (c) Add an extra floor to any one lift to reduce the total number of button presses by the maximum possible amount. What is the new number of button presses? [5]

3. Consider the set $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

   $i \neq j$

   (a) Construct eight distinct subsets $T_1, \ldots, T_8$ of S, such that $1 \leq i \leq 8, |T_i \cap T_j| = 7$. [4]

   i, j

   (b) Construct nine distinct subsets $T_1, \ldots, T_9$ of S, such that, for any i of these subsets, the common intersection has cardinality $9 - i$. [4]

4. Consider the recurrence relation: $a_n = 2a_{n-1} - a_{n-2}$ for $n \geq 2$.

   (a) Is it possible that $a_{5000} = 5000$ and $a_{10000} = 5100$? Prove or disprove. [3]

   (b) For the above recurrence relation, prove or disprove that $a_{n+1} > a_n, \forall n \geq 2$ if and only if $a_1 > a_0$ [4]

5. Let $S = \{s_1, s_2, \ldots, s_8\}$ be a set of 8 students, and let $G = \{g_1, g_2 \ldots, g_8\}$ be the set of 8 grades to be assigned to students from the set S. Find out the number of ways of assigning grades to these students that satisfy **all** of the following constraints. [4]

   (a) At least two students should get the same grade.

   (b) At least one grade should not get assigned to any student.

   (c) The student s_1 should not be given a g_1 grade.

   (d) The student s_2 should not be given a g_2 grade.

   Note that this is a single question, and not a question with four parts.

6. Chinese Remainder Theorem

   (a) Solve the congruence relations: $x \equiv 7(\mod 15)$ and $x \equiv 4(\mod 11)$ [2]

   (b) Let x and y be integers such that $x \equiv 5(\mod 15)$, $x \equiv 2(\mod 11)$, $y \equiv 3(\mod 15)$, and $y \equiv 8(\mod 11)$. Find $xy(\mod 15)$ and $xy(\mod 11)$. [3]

   (c) Find $a_n(\mod n)$, where $a_n = \sum_{k=1}^{n} k^3$, for any $n \geq 1$. You may not use a known formula for the sum of the first n cubes, without deriving it. [4]

7. Give your honest feedback on any aspect of the course. Irrespective of your feedback, and whether you attempt it or not, you will be awarded 4 marks for this question. So please be frank. [4]

T T   T V $(T,T) \Rightarrow T$

F T   F T $(F,T) \Rightarrow T$

F F   F T $(F,T) \Rightarrow T$

      F F $(F,F) \Rightarrow T$